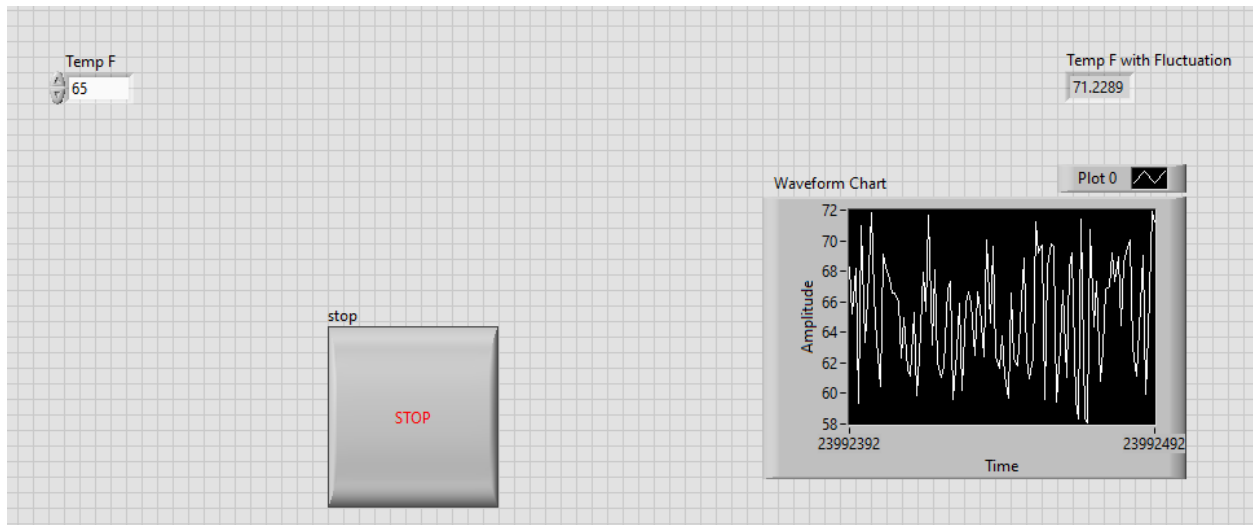


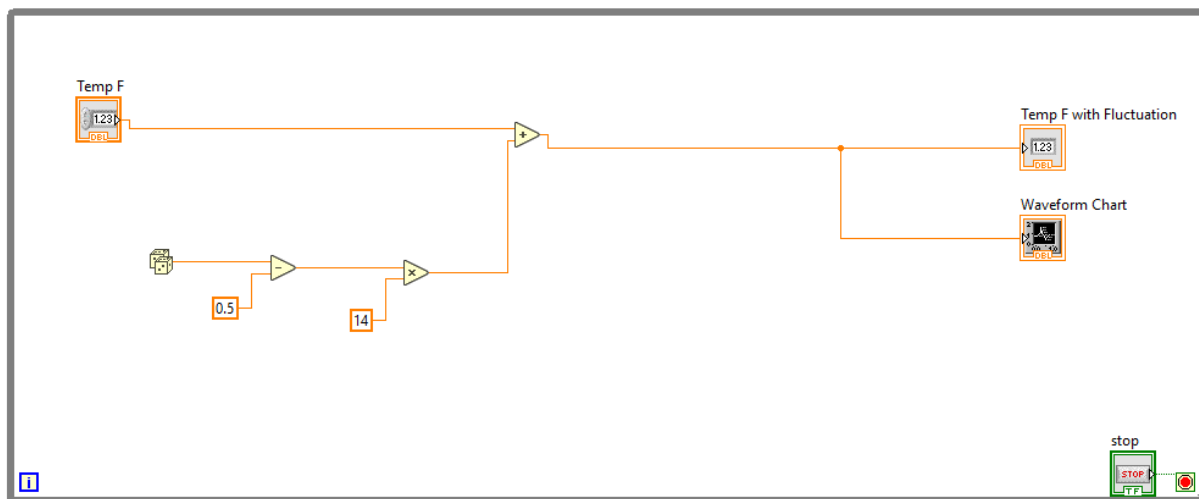
## Part 1

Front Panel:



I initially started with a numeric controller and a numeric indicator. I chose 65 for the value of the controller as the value required.

Block Diagram:

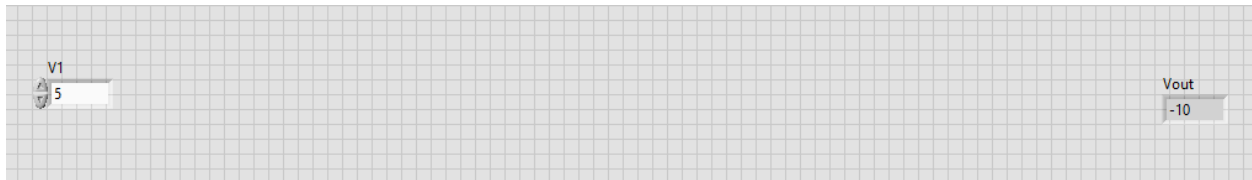


I worked out on paper how I would do the random fluctuation. I knew I could use the random number (dice) to generate a num between 0 and 1. Then I wanted to change the range, so I subtracted 0.5 from it and changed my range to  $[-0.5, 0.5]$  which was more useful since both sides were the same number acting as  $\pm$ . Then I just had to multiply get 7, so I multiplied by 14. This gave me a num with range  $[-7, 7]$  as needed.

It specified to run as long as needed, so added a while loop and associated stop button. I realized I would better see the results with a graph, so I added a waveform chart as well to show that it had an average of 65 while going to a max of 72 (+7) and a min of 58 (-7).

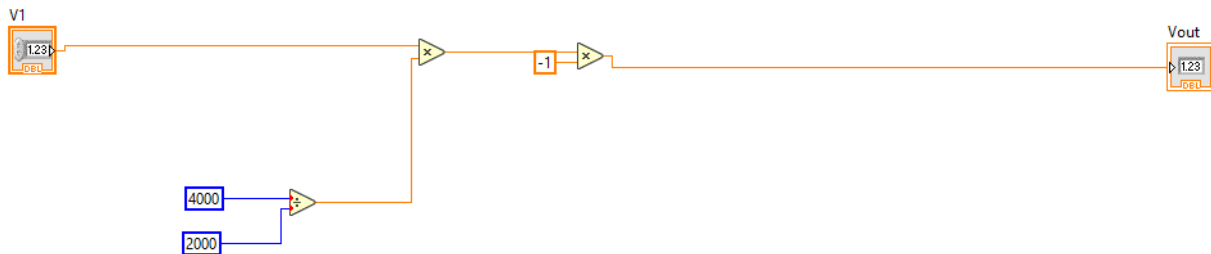
## Part 2a

Front Panel:



I started with a numeric controller for  $V_1$  and a numeric indicator for  $V_{out}$  as those are the inputs and outputs the gain is the relationship between.

Block Diagram:

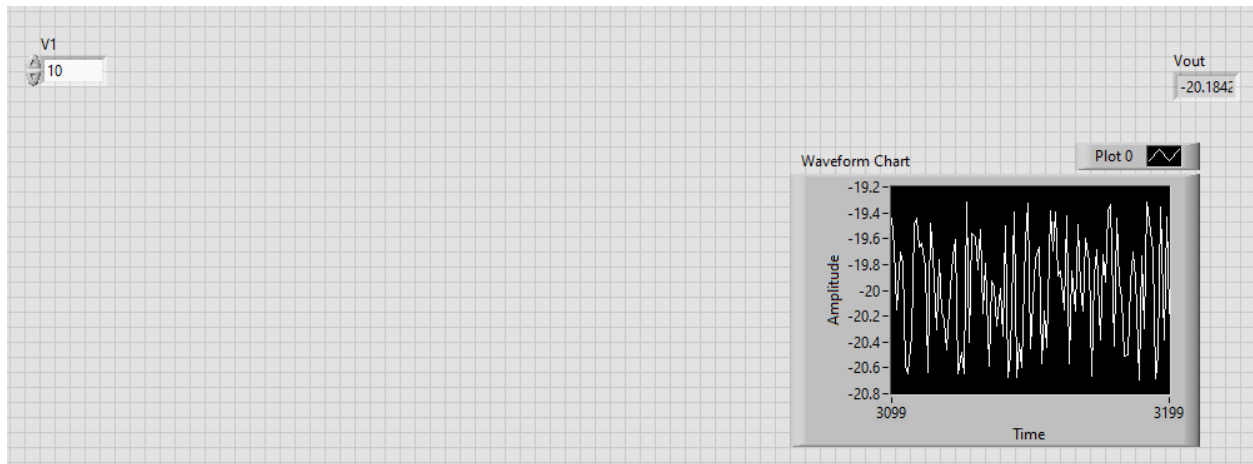


Finding the gain was relatively simple after recalling the equation for the gain of an inverting op-amp is  $-\frac{R_2}{R_1}$ . I created 2 numeric constants to symbolize  $R_2$  and  $R_1$  and had them divided, multiplied by  $V_1$ , then multiplied by -1 to symbolize the inverting.

I realized while going over my work that I should have displayed the gain rather than or in addition to  $V_{out}$ , but the gain is easily seen to be -2 when comparing  $V_1$  and  $V_{out}$ . You could display the gain itself by having the numeric constants do their dividing then multiply by -1 and display that value without multiplying by  $V_1$ .

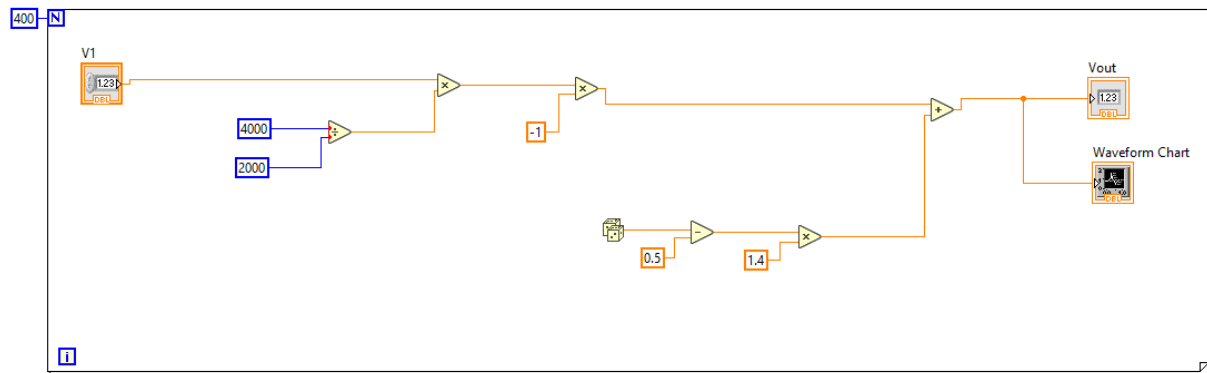
## Part 2b

Front Panel:



I kept a lot of the work front part 2a the same for part 2b, using the Save As feature to make a copy of my work from part 2a.

Block Diagram:



Then I added an additional random number generator to symbolize noise. I needed it to be  $\text{rnum} = [-0.7, 0.7]$  and it was currently  $\text{rnum} = [0, 1]$ , so I subtracted 0.5 to get me in a  $\pm$  state of  $[-0.5, 0.5]$  then multiplied by 1.4 to get  $[-0.7, 0.7]$ . This was a similar process to what was done in part 1.

There was also a requirement for running for 400 iterations, so I added a for loop and set N to run 400 times.

I added a waveform chart to display the output to an oscilloscope. The oscilloscope displayed the expected results.

I realized while going over this I forgot to set  $V_1$  to 3 V in my picture of the front panel, but the program should output the expected results for that as well.